



João Ventura

Master's in Distributed and Parallel Systems — NOVA FCT
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Profile

Second-year Master's student in Distributed and Parallel Systems at NOVA FCT, with a solid background in backend development, distributed architectures, and cloud infrastructure. Experienced in leading full-stack projects with real institutional partners, deploying self-hosted infrastructure, and working with HPC clusters. Currently exploring thesis directions in fault-tolerant distributed systems and edge computing. Seeking a summer internship or company thesis partnership for the 2025–2026 academic year.

Academic Background

Master's in Distributed and Parallel Systems 2025 – Present
NOVA FCT, Universidade NOVA de Lisboa — Second year in progress. Coursework: distributed algorithms, parallel computation, fault-tolerant architectures, cloud systems.

Bachelor's in Computer Science 2022 – 2026
NOVA FCT, Universidade NOVA de Lisboa — Final Grade: 14/20
Focus on distributed systems, software engineering, and functional programming. Coursework: Java, Python, Data Structures & Algorithms, AI, and Machine Learning. Tutored introductory programming and AI courses.

Professional Experience

Full-Stack Developer & Team Lead 2025
Land Management Platform — Academic & Municipal Partnership (NOVA FCT + Câmaras Municipais)
Led a 4-person team to design and deliver a full-stack web and mobile platform for sustainable land management in partnership with local municipal authorities. Enabled field operatives to collect georeferenced data, submit reports, and track operations in offline-capable conditions. Implemented GeoJSON mapping, GPX field tracking, real-time monitoring, JWT authentication, role-based access control, and reporting tools. Stack: Flutter, Spring Boot, PostgreSQL, Docker.

Programming Tutor 2021
TreeTree2
Invited to tutor following completion of TreeTree2's certified curriculum (Python, C++, algorithms, introductory AI, 2D game development). Delivered weekend sessions over two months teaching Python to young students, adapting to varying learning paces and tracking individual progress.

Technical Skills

Languages: Java, Python, C, JavaScript, Shell/Bash, Dart
Backend & Frameworks: Spring Boot, RESTful API Design, Microservices, Apache
Distributed Systems: Apache Kafka, Zookeeper, Paxos, Raft, event-driven architecture, fault-tolerant replication
HPC & Clusters: OAR (NOVA FCT DI-Cluster), parallel computation, distributed algorithm deployment
Databases: PostgreSQL, SQL, Firebase, Azure SQL
DevOps & Cloud: Docker, Git/GitHub, Google App Engine, Microsoft Azure, CI/CD (GitHub Actions)
Embedded & Hardware: ESP32, Arduino IDE **Mobile & Web:** Flutter, offline-capable mobile architecture
Other: Linux server administration, self-hosted infrastructure, system design, REST API testing (Postman)

Projects

Distributed Server–Client System 2025
Architected a distributed server–client system with fault-tolerant replication and strong consistency guarantees, implementing leader election via Zookeeper and event streaming via Apache Kafka.

Self-Hosted Server Infrastructure 2025
Configured and maintain a Linux home server running multiple personal services, covering system administration, network configuration, and end-to-end service deployment. Hosts a personal portfolio at jhonceda.duckdns.org (HTML, CSS, JavaScript).

Awards & Recognition

Honor Roll (2021): Awarded for exceptional dedication to supporting classmates; the only student recognised with this distinction that year.

Languages

Portuguese: Native **English:** Advanced (C2) **Spanish:** Intermediate (B1) **French:** Beginner (A2)

Hobbies

Cooking, music (listening, composing, and playing), hardware tinkering, and creating art.